Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/22/2005
PI Name:	Moore, Steven T. Ph.D.		
Project Title:	Validation of Centrifugation as a Countermeasure for Otolith Deconditioning During Spaceflight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	s.moore@cqu.edu.au	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	212-241-1943
Organization Name:	Mount Sinai School of Medicine		
PI Address 1:	Human Aerospace Laboratory		
PI Address 2:	Department of Neurology		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10029	**Congressional District:**	14
Comments:	NOTE: PI moved to Central Queensland University, Australia, July 2016.		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	07/02/2001	**End Date:**	07/01/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	Pre/Post Flight		
Flight Assignment:	NOTE: Changed end date to 7/1/2005 since was deselected (per PI 7/05 and S. McCollum 12/06). STS-118		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Bernard (Mount Sinai School of Medicine) Kaufmann, Horacio (Mount Sinai School of Medicine) Diedrich, Andre (Vanderbilt University) Biaggioni, Italo (Vanderbilt University) Clement, Gilles (CNRS Toulouse France)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In contrast to previous studies, post-flight measures of both otolith-ocular function and orthostatic tolerance were unimpaired in four payload crewmembers exposed to artificial gravity generated by in-flight centrifugation (Fig. 1B) during the Neurolab (STS-90) mission. The aim of the current proposal is to obtain control measures of otolith and orthostatic function following short duration missions, utilizing the centrifugation and autonomic testing techniques developed for the Neurolab mission, from astronauts who have not been exposed to in-flight centrifugation. This will enable a direct comparison with data obtained from the Neurolab crew. Deficits in otolith-ocular reflexes would support the hypothesis that intermittent exposure to in-flight centripetal acceleration is a countermeasure for otolith deconditioning. Furthermore, a correlation between post-flight otolith deconditioning and orthostatic intolerance would establish an otolithic basis for this condition. Obtain evidence for the efficacy of short-radius intermittent artificial gravity as a countermeasure to otolith and cardiovascular deconditioning on orbit. If proved effective, AG may be critical to maintaining crew health during future long-duration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this study are also relevant to studies of imbalance and orthostatic intolerance on Earth. Many of the postural and locomotor deficits observed in astronauts post-flight are similar to those seen in patients with vestibular disease, and findings from this study may shed light on the otolithic basis for these conditions.
Task Progress:	This project was deselected in December 2003.
Bibliography Type:	Description: (Last Updated: 09/07/2020)
Abstracts for Journals and Proceedings	Moore ST, MacDougall H, Cohen B, Wuyts F, Lesceu X, Speyer J-J, Clark J. "Head-eye coordination during simulated Orbiter landings." Aerospace Medical Association (AsMA) Scientific Meeting, Anchorage, AK, May 2004. Proceedings Aerospace Medical Association (AsMA) Scientific Meeting, Anchorage, AK, May 2004. , May-2004
Abstracts for Journals and Proceedings	MacDougall H, Moore ST, Cohen B. "Ambulatory monitoring of high frequency otolith stimulation." Society for Neuroscience Annual Meeting, San Diego, CA, November 2004. Proceedings Society for Neuroscience Annual Meeting, San Diego, CA, November 2004. , Nov-2004
Articles in Peer-reviewed Journals	Zhu D, Moore ST, Raphan T. "Robust and real-time torsional eye position calculation using a template-matching technique." Comput Methods Programs Biomed. 2004 Jun;74(3):201-9. PMID: 15135571 , Jun-2004
Articles in Peer-reviewed Journals	Moore ST, Cohen B, Raphan T, Berthoz A, Clement G. "Spatial orientation of optokinetic nystagmus and ocular pursuit during orbital space flight." Exp Brain Res. 2005 Jan;160(1):38-59. PMID: 15289967 , Jan-2005
Articles in Peer-reviewed Journals	Moore ST, Diedrich A, Biaggioni I, Kaufmann H, Raphan T, Cohen B. "Artificial gravity: a possible countermeasure for post-flight orthostatic intolerance." Acta Astronaut. 2005 May-Jun;56(9-12):867-76. PMID: 15835033 , May-2005